

Experiment 'p_6_2_c_replace_even_with_zero_seq' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_c_replace_even_with_zero_seq

Natural language statement: Write a method to replace all even elements in an array with 0.

Method signature: p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `forall i :: 0 <= i < |inputs| ==> (inputs[i] % 2 == 0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to replace all even elements in an array with 0.

The signature should be:

`p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

`ensures |result| == |inputs|, ensures forall i :: 0 <= i < |inputs| ==> (inputs[i] % 2 ==`
 ↪ `0 ==> result[i] == 0) && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0DxDxuht3CDoirz1pfchTNjI70L

Model: gpt-5.5-2026-04-23

Created at: 1780571305

Finish reason: stop

Usage: 374 tokens in, and 1079 tokens out

```
{  
  result := seq(|inputs|, i => if inputs[i] % 2 == 0 then 0 else inputs[i]);  
}
```

Final Program

```
method p_6_2_c_replace_even_with_zero_seq(inputs: seq<int>) returns (result: seq<int>)  
  ensures |result| == |inputs|  
  ensures forall i :: 0 <= i < |inputs| ==> (inputs[i] % 2 == 0 ==> result[i] == 0)  
    ↪ && (inputs[i] % 2 != 0 ==> result[i] == inputs[i])  
{  
  result := seq(|inputs|, i requires 0 <= i < |inputs| => if inputs[i] % 2 == 0 then 0  
    ↪ else inputs[i]);  
}
```

Total Token Usage

Input tokens: 374

Output tokens: 1079

Reasoning tokens: 1024

Sum of 'total tokens': 1453

Experiment Timings

Overall Experiment started at 1780571304905, ended at 1780571349681, lasting 44776ms (44.78 seconds)

Iteration #1 started at 1780571304905, ended at 1780571349681, lasting 44776ms (44.78 seconds)